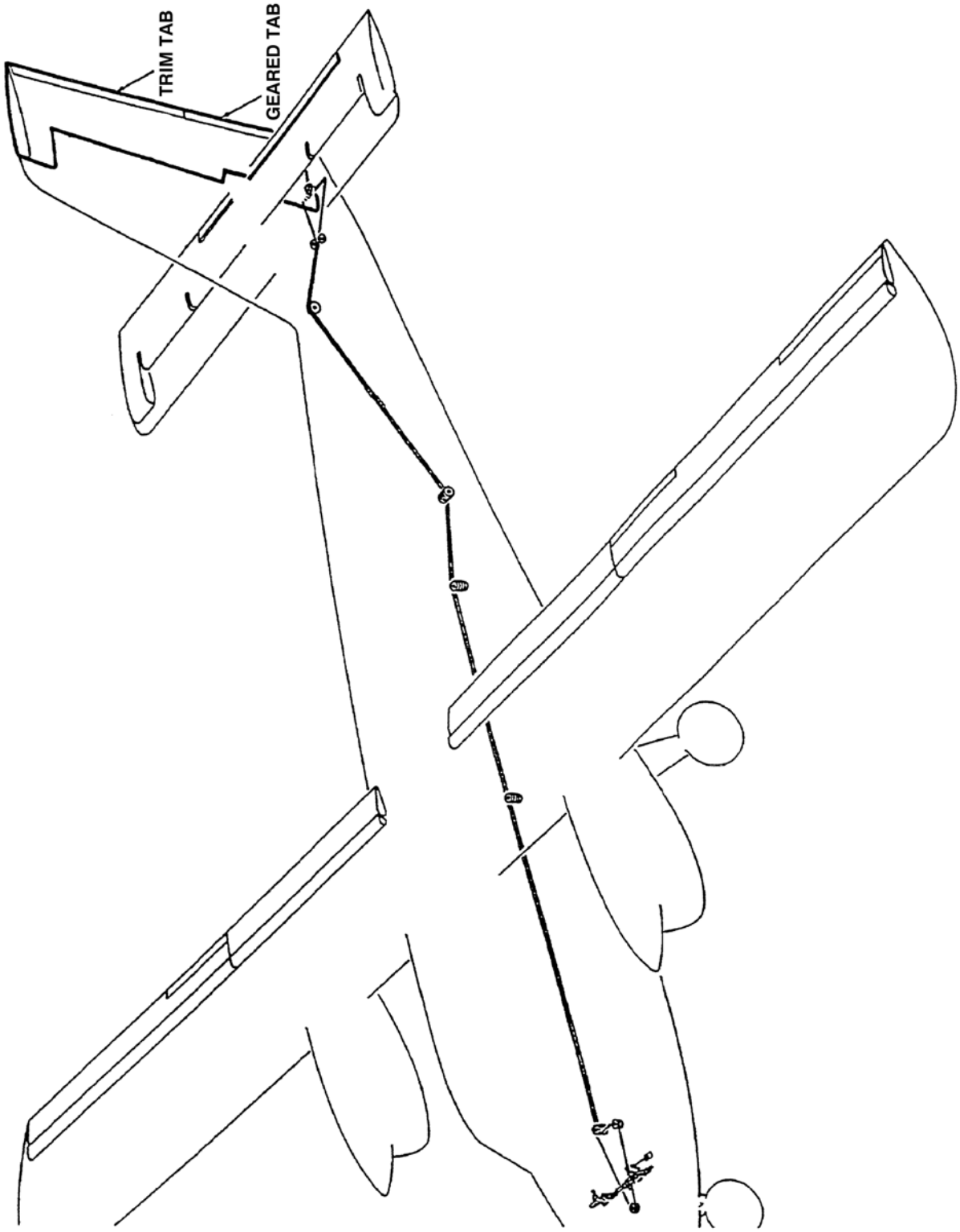
Rudder Control System – General Data

1. General

The rudder system provides operation of the rudder from the pilot's and co-pilot's positions. It consists of two rudder pedal assemblies, a torque tube, a rudder control quadrant, rudder lever, and rudder. The rudder pedals transmit the movement through the torque tube and rudder control quadrant located under the flight compartment floor. From the quadrant, two cables convey the movement to the rudder lever mounted on the rudder. The rudder cables are routed along the right side of the aircraft using the same banks of pulleys as for the elevator system, then change direction into the center of the rear fuselage. The rudder is hinged to the vertical stabilizer and rear fuselage at three points. Left and right rudder travel is limited by rudder stops, but, because left and right travel is unequal, the stops are of a different size. As an aid to identification the left stop is coloured blue, and the right stop grey. Two tabs are hinged to the rudder trailing spar, the upper being the trim tab, and the lower the geared tab (refer to 27-23-00, Rudder Trim Tab System – General Data and 27-26-00, Rudder Geared Tab System – General Data). Refer to Figure 1 for details.

Note For general cable rigging instructions, refer to 27-00-00, Flight Controls – General Data, Para 2.A.



Rudder System
Figure 1

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2. Adjustment/Test

A. Rig Rudder Control System

Refer to Figure 2 for details.

1) Set rudder pedals to highest position by pulling out adjustment catch, and engaging in bottom hole.

2) Engage rudder gust lock.

3) Using scale SD12568, check rudder neutral position by measuring 18.73 ± 0.06 inches between centers of index screw and bolt in rudder lever (dimension L, Figure 2).

4) If measurement is not within limits, adjust turnbuckles in rear fuselage as necessary to obtain dimension; tension cables to correct values. (Refer to 27-00-00, Flight Controls – General Data).

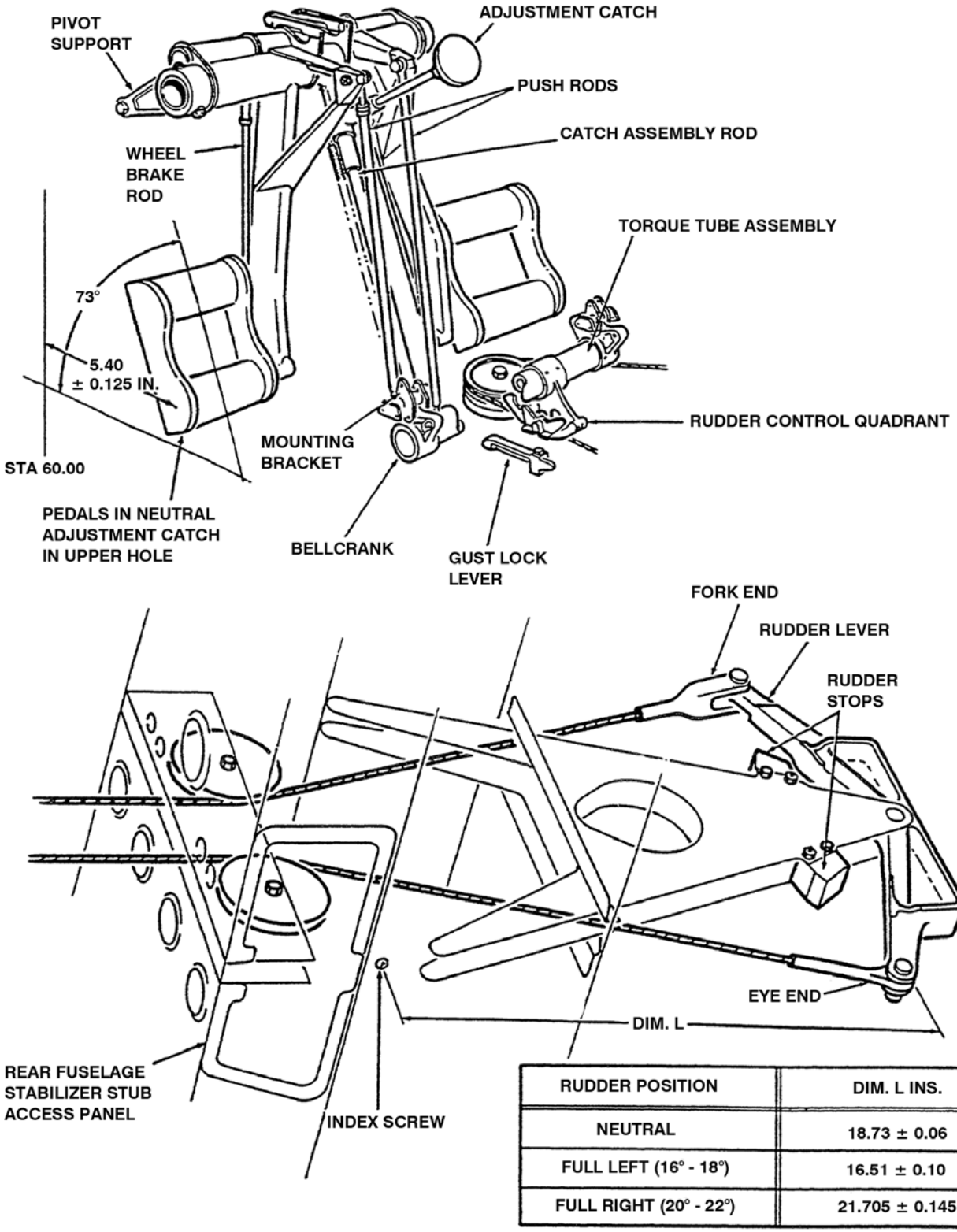
5) Set rudder pedal catch in highest hole, check with rudder in neutral, pedals align and angle of pedals is 73°, with a measurement of 5.40 ± 0.125 inches between bulkhead at station 60.00 and rudder pedal center pivots. Adjust push rods as necessary. Tighten nuts if disturbed.

6) Disengage rudder gust lock. Lock turnbuckles using locking clips.

7) Using scale SD12568, check dimension L (Figure 2) for full left and right rudder positions.

8) Operate rudder and check for full and free range of movement.

9) Repeat Para 2.A.7) and Para 2.A.8) with rudder pedals adjusted to each of the other three positions.



RUDDER POSITION	DIM. L INS.
NEUTRAL	18.73 ± 0.06
FULL LEFT (16° - 18°)	16.51 ± 0.10
FULL RIGHT (20° - 22°)	21.705 ± 0.145

Rudder System Rigging
Figure 2

3. Inspection/Check

A. Job Set-up

1) Either pull and clip breakers required to isolate affected circuits or placard and select aircraft electrical power to OFF.

2) Obey all relevant WARNINGS and CAUTIONS detailed in Maintenance Manual PSM 1–64–2, Chapters 12 and 24.

3) Placard flight controls in flight compartment.

B. Flight Compartment

1) Open access doors 15 to gain access to rudder cables (refer to Table 1 — Rudder Cable Assy - Flight Compartment and Passenger Cabin and Figure 2).

2) Do a visual inspection of control cables (non-jacketed) visible at this section of fuselage (refer to Table 1 — Rudder Cable Assy - Flight Compartment and Passenger Cabin and Figure 2). Observe carefully areas where cables pass over pulleys, quadrants, drums, or through fairleads.

3) Replace cables not acceptable to criteria in Inspection of Control Cables (refer to 27-00-00, Flight Controls – General Data). Make sure the new cable part numbers are identical to the discarded cable part numbers (refer to Table 1 — Rudder Cable Assy - Flight Compartment and Passenger Cabin).

4) Lubricate new or existing non-jacketed Carbon Steel cables (refer to 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data).

Note Do not lubricate stainless steel cables.

5) Replace cable as required.

6) Install and rig replaced cable assemblies (refer to 27-00-00, Flight Controls – General Data).

7) Close access doors 15.

Table 1 Rudder Cable Assy - Flight Compartment and Passenger Cabin

		CARBON STEEL	STAINLESS STEEL
Rudder quadrant inboard sheave to station 455.00	Pre Mod 6/2483	NAS305–34–4201	NAS305R34–4201
	Post Mod 6/2483	C6CF1511–3	C6CF1511–7
Rudder quadrant outboard sheave to station 455.00	Pre Mod 6/2483	NAS305–35–4266	NAS305R35–4266
	Post Mod 6/2483	C6CF1511–1	C6CF1511–5

C. Passenger Cabin

1) Remove panel 43 (refer to Figure 1, 12-00-00, Servicing – General Data) to gain access to rudder cables and pulleys (refer to Table 1 — Rudder Cable Assy - Flight Compartment and Passenger Cabin and Figure 2).

2) Release, attach string and pull back rudder cables to allow visual inspection (refer to Table 1 — Rudder Cable Assy - Flight Compartment and Passenger Cabin, Figure 2 and 27-00-00, Flight Controls – General Data Para 3.A).

3) Observe carefully areas where cables pass over pulleys, quadrants, drums, or through fairleads.

4) Replace cables not acceptable to criteria in Inspection of Control Cables (refer to 27-00-00, Flight Controls – General Data). Make sure the new cable part numbers are identical to the discarded cable part numbers per Table 1 — Rudder Cable Assy - Flight Compartment and Passenger Cabin.

5) Lubricate new or existing non-jacketed Carbon Steel cables (refer to 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data).

Note Do not lubricate stainless steel cables.

6) Install and rig cable assemblies (refer to 27-00-00, Flight Controls – General Data).

7) Install panel 43.

D. Rear Fuselage and Tail Area

CAUTION **WHEN INSTALLING ELEVATOR AND RUDDER PRIMARY FLIGHT CONTROL CABLES, CARGO AREA FLOOR BOARDS (BETWEEN STA 332.0 TO 376.0) AND AFT CARGO AREA FLOOR BOARDS (AFT OF STA 376.0), MUST BE REMOVED TO ENSURE PROPER INSTALLATION. THIS WILL REDUCE THE POSSIBILITY OF MISSRIGGING AND CABLE CROSSING, WHICH COULD LEAD TO ACCELERATED CABLE WEAR TO THE ELEVATOR AND RUDDER PRIMARY FLIGHT CONTROL SYSTEMS.**

1) Remove floor panels to gain access to rudder cable assemblies in baggage compartment rear fuselage.

2) Remove panels 11 and 12 (refer to 12-00-00, Servicing – General Data, Figure 1) to gain access to rudder cable assemblies in the tail area.

3) Do a visual inspection of control cables (non-jacketed) visible at this section of fuselage (refer to Table 2 — Rudder Cable Assy – Rear Fuselage and Tail Area, Figure 2 and 27-00-00, Flight Controls – General Data Para 3.A). Observe carefully areas where cables pass over pulleys, quadrants, drums, or through fairleads and cable ends. Place special emphasis on elevator and rudder cable inspection near cable cluster at station 376 (refer to Figure 2).

4) Replace cables not acceptable to criteria in Inspection of Control Cables (refer to 27-00-00, Flight Controls – General Data). Make sure the new cable part numbers are identical to the discarded cable part numbers per Table 2 — Rudder Cable Assy – Rear Fuselage and Tail Area.

5) Lubricate new or existing non-jacketed Carbon Steel cables only (refer to 20-60-01, Inspection, Cleaning and Lubrication of 7x19 and 7x7 Control Cables – General Data).

Note Do not lubricate stainless steel cables.

6) Install and rig replaced cable assemblies (refer to 27-00-00, Flight Controls – General Data).

7) Install baggage compartment floor panels and panels 11 and 12.

Table 2 Rudder Cable Assy – Rear Fuselage and Tail Area

	CARBON STEEL	STAINLESS STEEL
Station 455.0 to right-hand side rudder control attachment	C6CF1150–1	E.O. 69053–6
Station 455.0 to left-hand side rudder control attachment	C6CF1151–1	E.O. 69053–7

E. Close-out:

1) Remove placards on the flight controls in flight compartment.

2) Obey all relevant WARNINGS and CAUTIONS detailed in Maintenance Manual PSM 1–64–2, Chapters 12 and 24.

3) Either remove clip and push breakers that were required to isolate affected circuits or remove placards and select aircraft electrical power to ON.

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